Calibration of Compactors

A number of manufacturers produce Superpave Gyratory Compactors. See Figure 4.15 for a composite photo of different compaction equipment.

In the fabrication of the SGC specimens, a constant consolidation pressure is applied to the sample while the mold gyrates at a nominally constant angle (the angle of gyration). To provide specimens of consistent density, it is very important that the SCC maintain a constant pressure and a known constant angle of gyration during the compaction process. Each model of SGC uses a different method of setting, inducing, and maintaining the angle of gyration. Each model also uses a unique calibration system to measure the angle of gyration externally in relation to the mold. These calibration systems cannot be used universally on all models of SGCs. Inconsistencies in the density of asphalt specimens produced by different SGCs can be attributed to variations in the angle of gyration. This can be very troublesome when comparing mix designs prepared by different compactors or during quality control, when the quality control specimens are manufactured using one brand of compactor and the acceptance specimens are manufactured using another brand of compactor. The American Association of State Highway and Transportation Officials (AASHTO) have developed a procedure for evaluating the internal angle of a SGC. It is AASHTO PP 48 "Evaluation of the Superpave Gyratory Compactor (SGC) Internal Angle of Gyration."

Specimen Preparation and Compaction

The following procedure outlines the procedure for the preparation of HMA test specimens using the SGC. Detailed steps for the preparation of the specimens are included in the SP-2 manual.

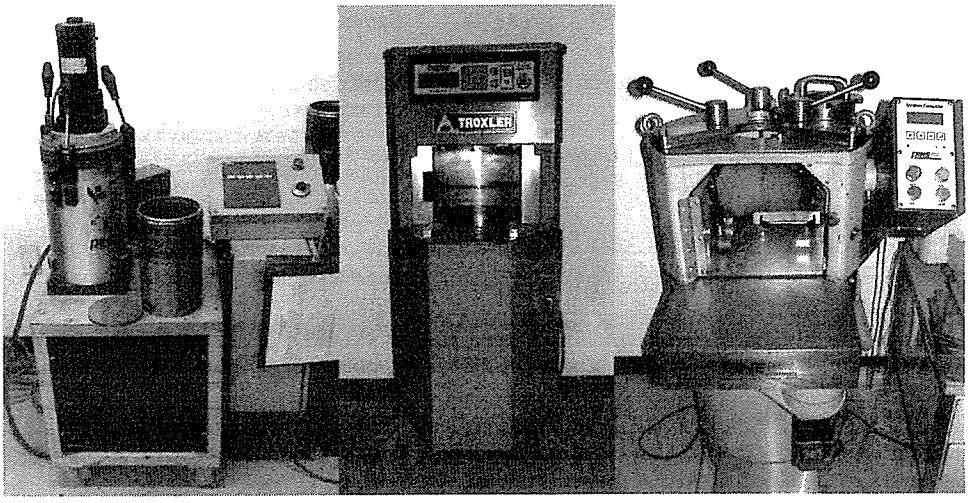


Figure 4.15 Composite Photo of Gyratory Compactors

Preparation of Aggregate

Each aggregate blend being evaluated should have a batch sheet showing the batch weights of each aggregate component and the asphalt binder. Three mix sample sizes are needed to complete a mix design. For the compacted mix specimens that will be used in the Superpave mix design, the specimen size is 150 millimeters (diameter) by 115 millimeters (height) and requires approximately 4700 grams of aggregate. Samples of mix to be used for the determination of the maximum specific gravity using AASHTO T 209 or ASTM D 2401 will remain uncompacted, and their size will vary by aggregate size, ranging from 1000 grams to 2500 grams. Moisture testing using AASHTO T 283 or ASTM D 4867 requires a mix specimen height of 95 millimeters and will require approximately 3700 grams of aggregate.

Compaction of Volumetric Specimens

Detailed procedures for the preparation of the volumetric specimens are contained in the SP-2 manual. The specimens are compacted to either N_{des} or N_{max} as required by the specifying agency. A minimum of two specimens are compacted at the trial blend's estimated asphalt content, at 0.5 percent of the estimated asphalt content, and at 1.0 percent above the estimated asphalt content. Four asphalt contents are required for the accomplishment of a Superpave mix design. Many designers will estimate the trial binder content based on experience.

The specimens are mixed and compacted at temperatures using a plot of viscosity versus temperature (Figure 4.16). This data is available from the asphalt binder supplier. The mixing and compaction temperatures are those corresponding to a binder viscosity value of 0.17 ± 0.02 Pascal-second and 0.28 ± 0.03 Pascal-second, respectively. These viscosity ranges are not valid for modified asphalt binders. The designer should consider the manufacturer's recommendations when establishing mixing and compaction temperatures for modified binders. Practically, the mixing temperature should not exceed

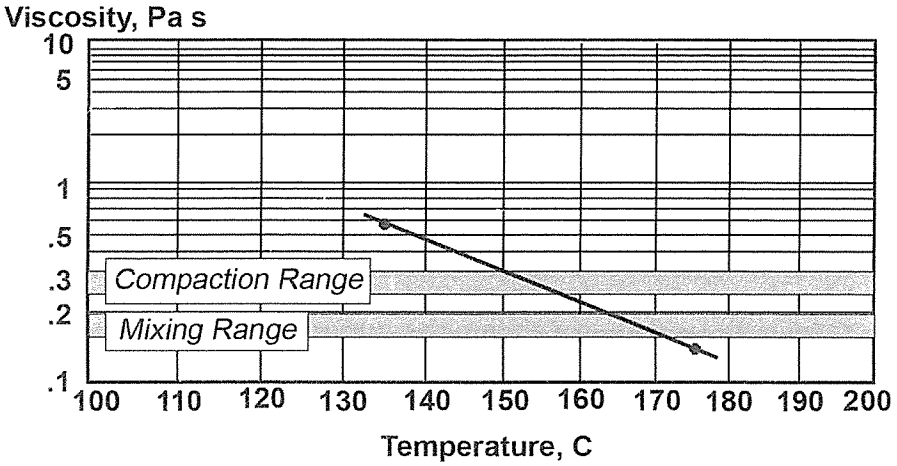


Figure 4.16 Temperature-Viscosity Relationship

325°F (165°C) and the compaction temperature should not be lower than 240°F (115°C).

Superpave mix design requires that all samples, prior to compaction, must be mixed and oven-aged at compaction temperature for a minimum of 2 hours. This is required to allow absorption of the asphalt binder into the aggregates before compaction. Field experience has shown that aggregates with more than 2 percent water absorption should be aged for 4 hours to allow additional absorption to occur. Specimen aging simulates the absorption that occurs during the mixing and placement process for the HMA. If the mix is not aged, a lower asphalt content may be selected, which could result in some loss in durability during the life of the pavement.

Data Analysis and Presentation

After compaction of the samples, the volumetric properties of each specimen are determined. The results of the laboratory testing are plotted to provide curves of asphalt content versus air-voids, voids mineral aggregate, and voids filled with asphalt. Typical curves are shown in Figure 4.17.

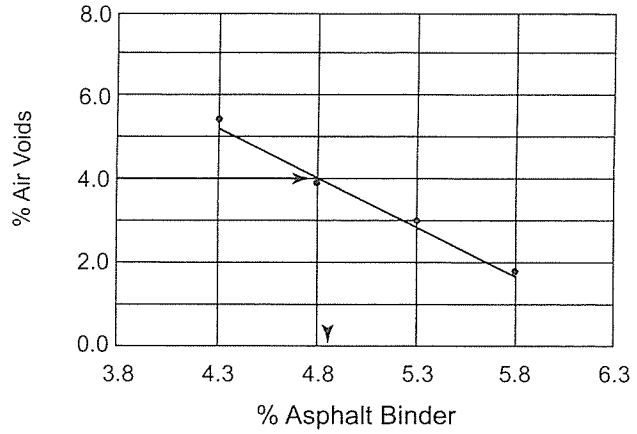
4.3.4 Design Asphalt Binder Content

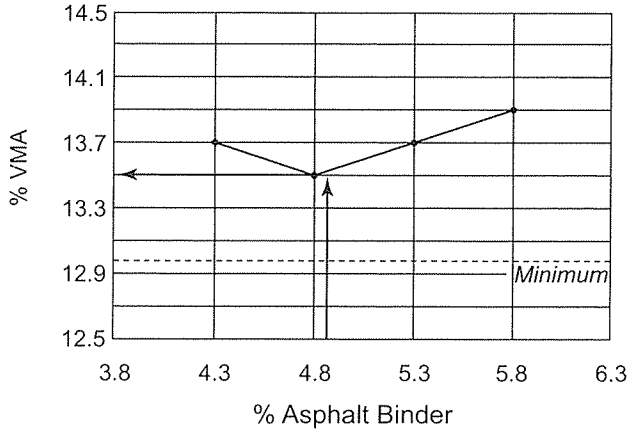
After the data has been plotted, the designer will pick an asphalt binder content that will provide 4 percent air-voids at N_{des}. Then the designer determines the values for voids in the mineral aggregate (VMA) and voids filled with asphalt (VFA) from the other curves. These values will then be checked against the mixture specifications. The recommended Superpave volumetric mixture design requirements are shown in Table 4.5.

4.3.5 Steps That Can Be Taken to Adjust Properties

If the testing indicates that the mix does not meet the mix design requirements, the following suggestions are provided (AASHTO R 35):

- **Adjusting the VMA**—A change in the aggregate gradation or aggregate source is usually required to adjust the VMA. There are three likely options:
  - Change the gradation. Moving the gradation curve toward the maximum density line will generally reduce the VMA; moving it away from the maximum density line will generally increase the VMA.
  - Change the amount of minus No. 200 (0.075 millimeters) material. Increasing the No. 200 material usually lowers the VMA, and decreasing it will typically raise the VMA.
  - Change the surface texture and/or the shape of the particles of the fine aggregate fraction. The increased use of an aggregate with a rough texture (higher FAA) will generally increase the VMA, and the increased use of a rounded aggregate will reduce the VMA.





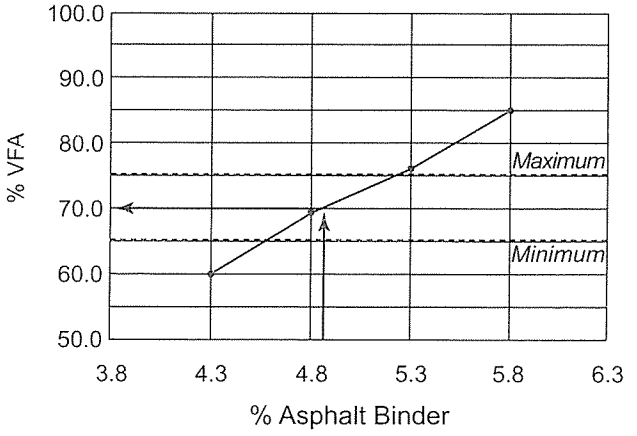


Figure 4.17 Typical Superpave Mix Design Curves

Table 4.5
Superpave Volumetric Mixture Design Requirements

Design ESALs (million)	Required Density (percent of Theoretical Maximum Specify Gravity)			Voids-in-the-Mineral Aggregate (percent), minimum					Voids Filled with Asphalt (percent)	Dust-to-Binder Ratio
				Nominal Maximum Aggregate Size, mm						
	$N_{initial}$	N_{design}	N_{max}	37.5	25.0	19.0	12.5	9.5		
< 0.3	≤ 91.5								70–80	
0.3 to < 3	≤ 90.5								65–78	
3 to < 10				11.0	12.0	13.0	14.0	15.0		
10 to < 30	≤ 89.0	96.0	≤ 98.0						65–75	0.6–1.2
≥ 30										

Design ESALs are the anticipated project traffic level expected on the design lane over a 20-year period. Regardless of the actual design life of the roadway, determine the design ESALs for 20 years and choose the appropriate N_{design} level.

For 9.5 millimeters nominal maximum size mixtures, the specified VFA range shall be 73 to 76 percent for design traffic levels ≥ 3 million ESALs.

For 25.0 millimeters nominal maximum size mixtures, the specified lower limit of the VFA range shall be 67 percent for design traffic levels < 0.3 million ESALs.

For 37.5 millimeters nominal maximum size mixtures, the specified lower limit of the VFA shall be 64 percent for all design traffic levels.

If the aggregate gradation passes beneath the boundaries of the aggregate restricted zone, consideration should be given to increasing the dust-to-binder ratio criteria from 0.6–1.2 to 0.8–1.6.

- **Adjusting the VFA**—The lower limit of the VFA range should always be met at 4.0 percent air-voids if the VMA meets the requirements. If the upper limit is exceeded, then the VMA is substantially above the minimum required. The result can be an HMA mixture that will not support heavy traffic. If this occurs, the steps discussed above should be taken to reduce the VMA.

4.3.6 Moisture Sensitivity

The loss of bond, or stripping, caused by the presence of moisture between the asphalt and the aggregate is a problem in some areas and can be severe in some cases. Many factors, such as aggregate characteristics, asphalt characteristics, environment, traffic, construction practices, and drainage, can contribute to stripping.

The moisture susceptibility test used to evaluate HMA for stripping is AASHTO T 283 or ASTM D 4867. This test is not a performance-based test, but it serves two purposes. First, it identifies whether a combination of asphalt binder and aggregate is moisture susceptible. Second, it measures the effectiveness of antistripping additives.

In the test, two subsets of test specimens are produced. Specimens are compacted to a height of 95 millimeters and to an air-void content of 6.5 to 7.5 percent with a target value of 7 percent. Test specimens should be sorted so that each subset has the same average air-void content. One subset is moisture conditioned by vacuum saturation to a constant degree of saturation in the range of 70 to 80 percent. This is followed by an optional freeze cycle. The final conditioning step is a hot-water soak. After conditioning, both subsets are tested for indirect tensile strength. After conditioning, the specimen is removed from the water bath and placed on its side between the bearing plates of the testing machine. Steel loading strips are placed between the specimen and the bearing plates. A load is applied to the specimen at the rate of 2 inches (50 millimeters) per minute. The maximum load is recorded. This is the indirect tensile strength of the specimen. This test is conducted on the conditioned and the unconditioned specimens. The average strength for the conditioned and the unconditioned specimens is determined. The ratio of these two values is the tensile-strength ratio or TSR. Superpave requires a minimum TSR of 80 percent. Lower values are indicative of mixtures that may exhibit stripping problems after construction.

Steps to Improve Moisture Resistance

The best way to improve the moisture resistance of an HMA mixture is a proper mix design with sufficient asphalt binder to coat the aggregates. This must be coupled with proper compaction in the field. The goal is to produce as impermeable an HMA mixture as possible to prevent the intrusion of water. However, some asphalt/aggregate combinations will require an additive to improve the moisture resistance of the mixture. The two most commonly used admixtures are hydrated lime and proprietary liquid antistrip agents. It is imperative that the mix design be accomplished using the admixture and the asphalt binder that will be used on the project. The reaction is chemical, and the results will vary depending on the type and amount of additive, source of aggregate and source of asphalt binder. For more information on moisture sensitivity, see the Asphalt Institute's MS-24, *Moisture Sensitivity* manual.

4.4 Marshall Method of Mix Design

4.4.1 Introduction and Historical Context

The concepts of the Marshall method of designing paving mixtures were formulated by Bruce Marshall, formerly a bituminous engineer with the Mississippi State Highway Department. The U.S. Corps of Engineers, through extensive research and correlation studies, improved and added certain features to Marshall's test procedure, and ultimately developed mix design criteria. The Marshall test procedures have been standardized by the American Society for Testing and Materials. Procedures are given by ASTM D 1559.

The Marshall method has generally been applicable to dense-graded HMA paving mixtures using Performance Graded (PG) asphalt binders and containing aggregates with a maximum size of 1 inch (25 millimeters) or less. It has also been adapted to the design of SMA mixtures. The Marshall method is used for both laboratory design and the quality control of hot mix asphalt pavements. For a detailed description of the steps and equipment needed to conduct a Marshall Mix Design, the reader is referred to the Asphalt Institute's MS-2 *Mix Design Methods* manual. The Asphalt Institute Mix Design Program SW-2 performs all of the calculations needed to conduct a Marshall Mix Design. (It is recommended that mix designers refer to either AASHTO or ASTM requirements, or to the design agency's requirements so that those specific procedures can be followed.)

4.4.2 Description of Method

The Marshall method uses standard cylindrical test specimens of 2½ inches (64 millimeters) in height by 4 inches (102 millimeters) in diameter. These are prepared using a specified procedure for heating, mixing, and compacting the asphalt-aggregate mixtures. These specimens are compacted using a Marshall compaction hammer (Figure 4.18) and tested in a Marshall stability and flow device (see Figure 4.19). The two principal features of the Marshall method of mix design are a density-voids analysis and a stability-flow test of the compacted test specimens.

The stability of the test specimens is the maximum load resistance in pounds (newtons) that the standard test specimen will develop at 140°F (60°C) when tested as outlined here. The flow value is the total movement or strain, in units of 1/100 inch (0.25 millimeter), occurring in the specimen between no load and maximum load during the stability test.

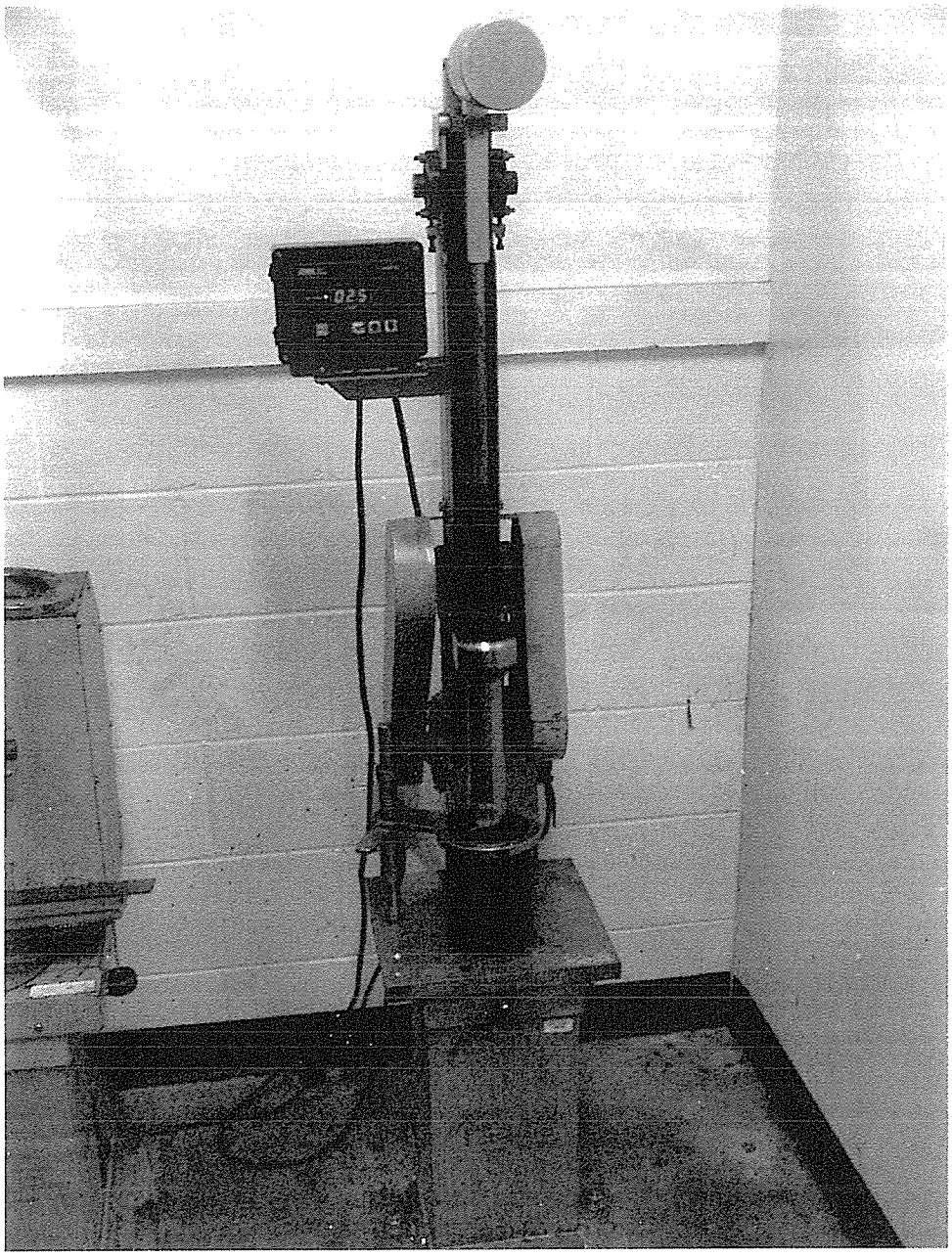


Figure 4.18 Marshall Compaction Hammer

The Marshall mix design process consists of three basic steps:

1. **Aggregate selection.** This step consists of insuring that the aggregates used in the mixture meet the agency's or the specifying engineer's requirements for the aggregates to be used in the mixture. For a detailed discussion of the aggregate requirements for an HMA mixture, the reader is referred to Chapter 3, "Mineral Aggregates and

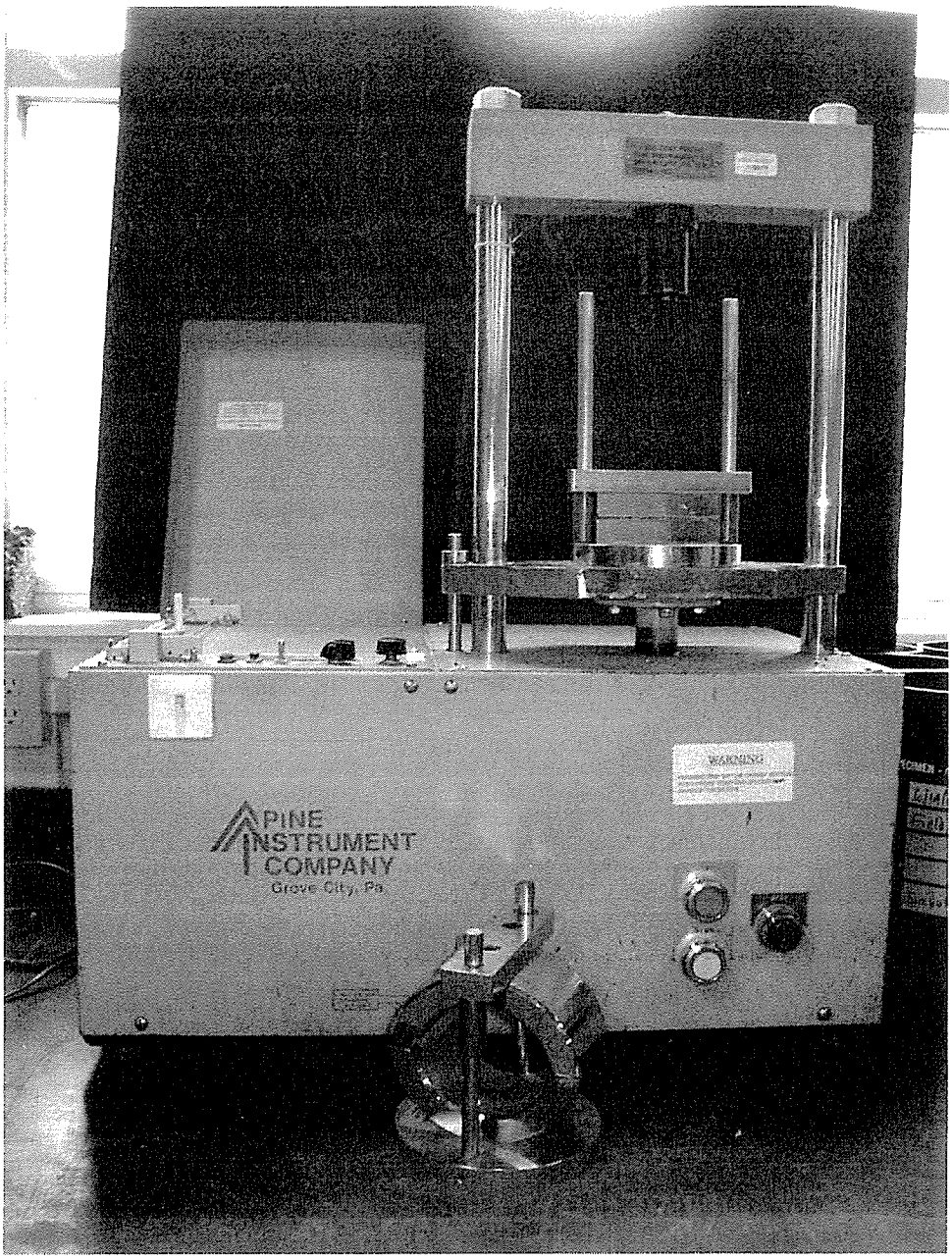


Figure 4.19 Marshall Stability and Flow Machine

Mixture Additives." Once the designer is assured that the aggregates meet the physical requirements specified, the next step is to blend those aggregates to meet the gradation and size requirements.

2. **Asphalt binder selection.** The designer identifies the asphalt binder to be used in the mixture based on the procedures described in Chapter 2, "Petroleum Asphalt."

3. **Manufacturing of test specimens with a particular blend or gradation of aggregates.** These test specimens are prepared for a range of different asphalt contents so that the test data curves show a well-defined "optimum" value. Tests are conducted on specimens that differ by 0.5 percent increments of asphalt content, with at least two asphalt contents above optimum and at least two below optimum. For example, for a mix where the estimated asphalt content is 4.5 percent, three samples each would be made at 3.5, 4.0, 4.5, 5.0, and 5.5 percent asphalt by weight for a total of 15 samples.
   a. These samples are compacted using the Marshall compaction hammer (Figure 4.18). Depending on the requirements of the agency, generally either 50 blows or 75 blows are used.
   b. The bulk specific gravity of each compacted sample is determined for use in the voids analysis.
   c. The samples are tested in the Marshall testing machine (Figure 4.19) for stability and flow. The Marshall stability and flow values are determined. The requirements for Marshall stability and flow will vary with the agency and the engineer.
4. A minimum of 2 additional uncompacted samples would be used for the determination of the maximum specific gravity as discussed in Section 4.2. The data from the maximum specific gravity test and the bulk specific gravity test are used, along with the specific gravities of the aggregates, to determine the volumetric properties of the mix. The data is plotted as shown in Figure 4.20.

The information shown in the plotted data is used to determine the optimum asphalt content for the mixture. The most common method is to determine the asphalt content that corresponds to 4 percent air-voids. That asphalt content is then used to determine the values for Marshall stability, flow, VMA, and VFA. For the example shown, the asphalt content would be 4.7 percent. Each of the remaining criteria are compared against the specification requirements. If they fall within the specification requirements, the asphalt content at 4 percent air-voids is the asphalt content chosen for production.

4.4.3 Mix Design Criteria

Whether the asphalt paving mix will be satisfactory from the design asphalt content selected is determined by applying certain limiting criteria to test data for the mix at its design asphalt content. An engineer designing an HMA mixture for an agency should check that agency's specific requirements. The design criteria in Table 4.6 are recommended by the Asphalt Institute. If all the criteria are met, then this is the design asphalt content. If any of the design criteria are not met, then some adjustment is necessary or the mix may need to be redesigned.

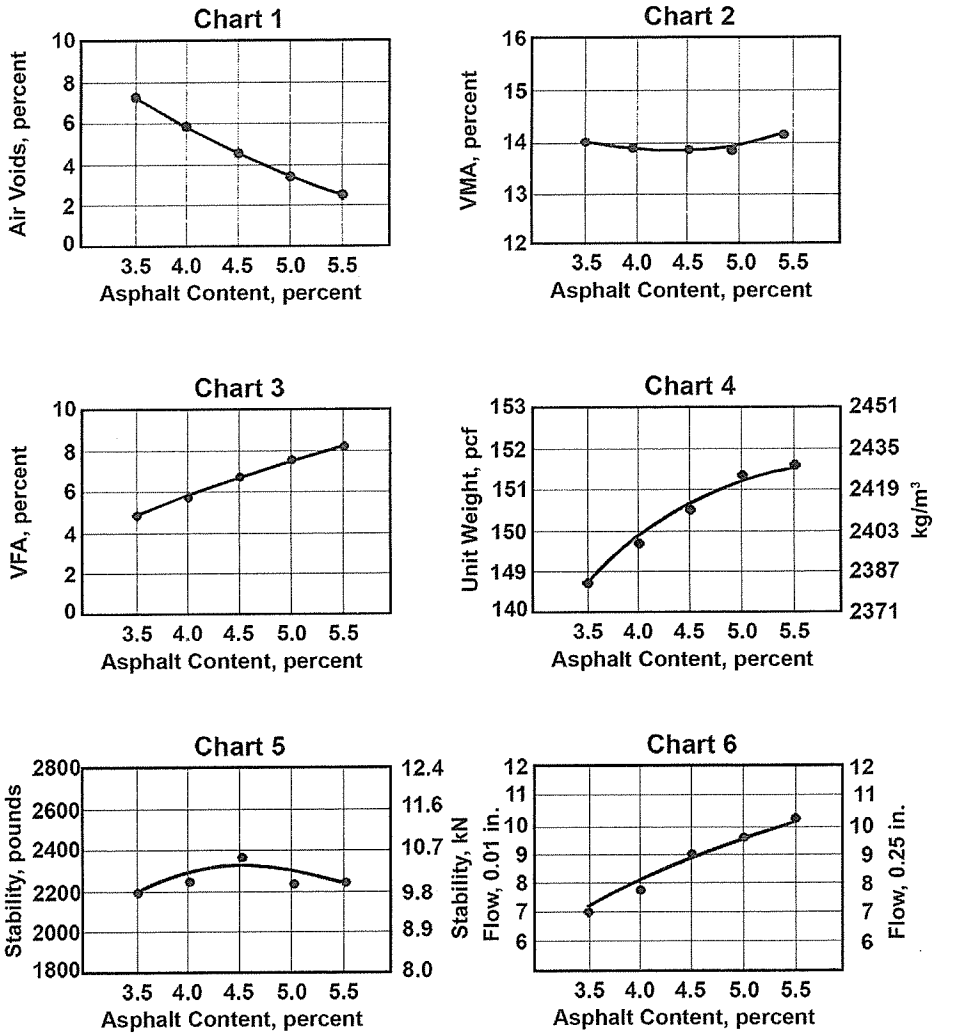


Figure 4.20 Test Properties for Hot Mix Design Data by the Marshall Method

The mix design selected for use in a pavement is usually the one that most economically meets all the established criteria. However, the mix should not be designed to optimize one particular property. For example, mixes with abnormally high values of stability are often less desirable because pavements with such mixes tend to be less durable and might crack prematurely under heavy volumes of traffic. Any variations in design criteria should be allowed only under unusual conditions, unless the service behavior of specific aggregate mixture indicates that such a variant paving mix is satisfactory.

Table 4.6
Marshall Design Criteria

Marshall Method	Light Traffic Surface & Base		Medium Traffic Surface & Base		Heavy Traffic Surface & Base	
Mix Criteria 1	Min.	Max.	Min.	Max.	Min.	Max.
Compaction, number of blows each end of specimen		35		50		75
Stability, lb.	750		1200		1800	
(N)	(3336)	–	(5338)	–	(8006)	–
Flow, 0.01 in (0.25 mm)	8	18	8	16	8	14
Percent Air-Voids	3	5	3	5	3	5
Percent Voids in Mineral Aggregate (VMA)			*See Figure 3.20*			
Percent Voids Filled With Asphalt (VFA)	70	80	65	78	65	75

Notes

1. All criteria, not just stability value alone, must be considered in designing an asphalt paving mix. Hot mix asphalt bases that do not meet these criteria when tested at 140°F (60°C) are satisfactory if they meet the criteria when tested at 100°F (38°C) and are placed 4 inches (100 millimeters) or more below the surface. This recommendation applies only to regions having a range of climatic conditions similar to those prevailing throughout most of the United States. A different lower test temperature may be considered in regions having more extreme climatic conditions.
2. Traffic classifications:
   Light—traffic conditions resulting in a Design ESAL < 10^4
   Medium—traffic conditions resulting in a Design ESAL between 10^4 and 10^6
   Heavy—traffic conditions resulting in a Design ESAL > 10^6
3. Laboratory compaction efforts should closely approach the maximum density obtained in the pavement under traffic.
4. The flow value refers to the point at which the load begins to decrease.
5. The portion of asphalt cement lost by absorption into the aggregate particles must be allowed for when calculating percent air-voids.
6. Percent voids in the mineral aggregate should be calculated on the basis of the ASTM bulk specific gravity for the aggregate.

4.5 Hveem Method of Mix Design

4.5.1 Introduction and Historical Context

The Hveem method of mix design is a laboratory procedure based on two properties of the compacted specimen: cohesion and friction. Density and voids of the compacted specimen are established before measuring the stability with the stabilometer. The stabilometer is a triaxial testing device in which vertical loads are applied and resulting lateral pressures are measured. The concepts of the Hveem method of designing paving mixtures were advanced and developed under the direction of Francis N. Hveem, who was the materials and research engineer for the California Division of Highways. The test procedures and their application were developed through extensive research and correlation studies on asphalt highway pavements.

The method shown is applicable to the design of hot, dense-graded asphalt paving mixtures.

4.5.2 Description of Method

Hveem method test procedures have been standardized by the ASTM and AASHTO. Test procedures are found in AASHTO T 246 or ASTM D 1560, and AASHTO T 247 or ASTM D 1561. Testing procedures presented here are basically the same as those in the ASTM procedure. For a detailed description of the Hveem procedure, the reader is referred to the Asphalt Institute's MS-2, *Mix Design Methods* manual. It is recommended that agencies refer to either AASHTO or ASTM, or to the design agency's requirements, so that those specific procedures can be followed.

The Hveem method uses standard test specimens of 2.5 inches (64 millimeters) in height by 4 inches (102 millimeters) in diameter; these are prepared using a specified procedure for heating, mixing, and compacting the asphalt-aggregate mixtures. The principal features of the Hveem method of mix design are the oil soak and Centrifuge Kerosene Equivalent (CKE) tests on the aggregates to estimate the asphalt requirements of the mix, followed by a stabilometer test, a cohesiometer test, a swell test, and a density-voids analysis on test specimens of the compacted paving mixtures. The stabilometer test uses a special triaxial-type testing cell for measuring the resistance of the compacted mix to lateral displacement under vertical loading, and the swell test measures the resistance of the mix to the action of water. The specimens are maintained at 140°F (60°C) for the stability tests; the swell tests are performed at room temperature.

The Hveem mix design process consists of three basic steps:

1. **Aggregate selection.** This step consists of insuring that the aggregates used in the mixture meet the design requirements. For a detailed discussion of the aggregate requirements for an HMA mixture, the reader is referred to Chapter 3, "Mineral Aggregates and Mixture Additives." Once the designer is assured that the aggregates meet the specified physical requirements, the next step is to blend the aggregates to meet the requirements.
2. **Asphalt binder selection.** The designer identifies the asphalt binder to be used in the mixture based on the procedures described in Chapter 2, "Petroleum Asphalt."
3. **Test specimen preparation.** A series of test specimens is prepared using a particular blend or gradation of aggregate over a range of different asphalt contents so that the test data curves show a well-defined optimum value. Tests should be scheduled on the basis of 0.5 percent increments of asphalt content, with at least two asphalt contents above optimum and at least two below optimum. The following procedure outlines the steps associated with the determination of the optimum asphalt content using the Hveem method:
   a. In the Hveem method, the initial asphalt content is determined by the oil soak and Centrifuge Kerosene Equivalent tests and the calculated surface area. For example, for a mix with an estimated asphalt content of 5.5 percent, one sample each would be made at 4.5, 5.0, 5.5, 6.0, and 6.5 percent asphalt by weight for a total of five samples. A minimum of 2 additional samples would be used for the determination of the maximum specific gravity as discussed in Section 4.2. Compact the samples using the California Kneading Compactor (see Figure 4.21).
   b. Test each sample for stability and cohesion using the Hveem Stabilometer and Cohesiometer (see Figure 4.22). The passing values of stability and cohesion are specific to the agency using the design procedure.
   c. Determine the volumetric properties of the mix. Data is plotted as shown in Figure 4.23.

4.5.3 Mix Design Criteria

The suitability of the hot mix design by the Hveem method is determined on the basis of whether the asphalt content and the aggregate grading will satisfy the requirements shown in Table 4.7. An engineer designing an HMA mixture for an agency should check that agency's specific requirements.



Figure 4.21 California Kneading Compactor

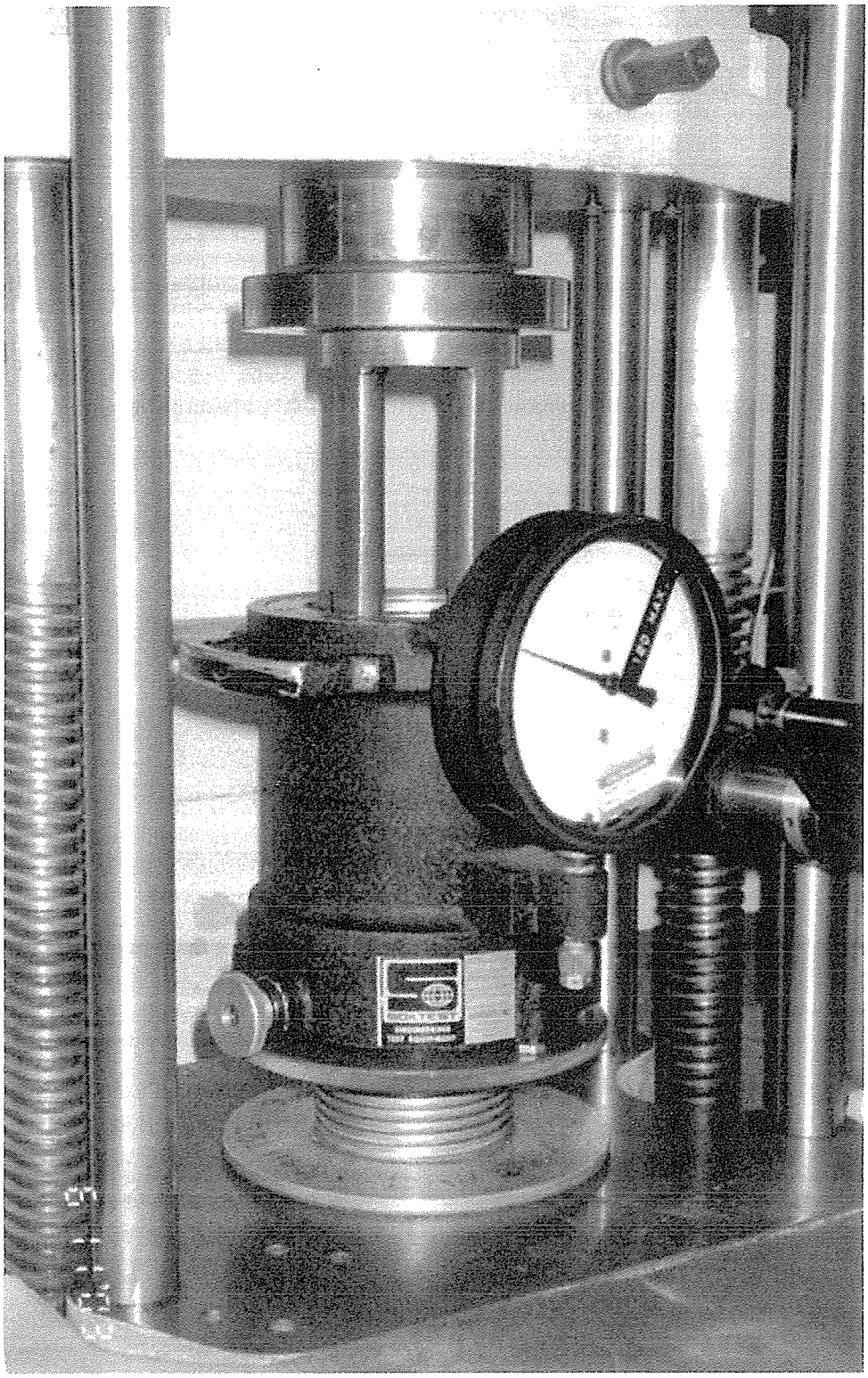


Figure 4.22 Hveem Stabilometer Machine

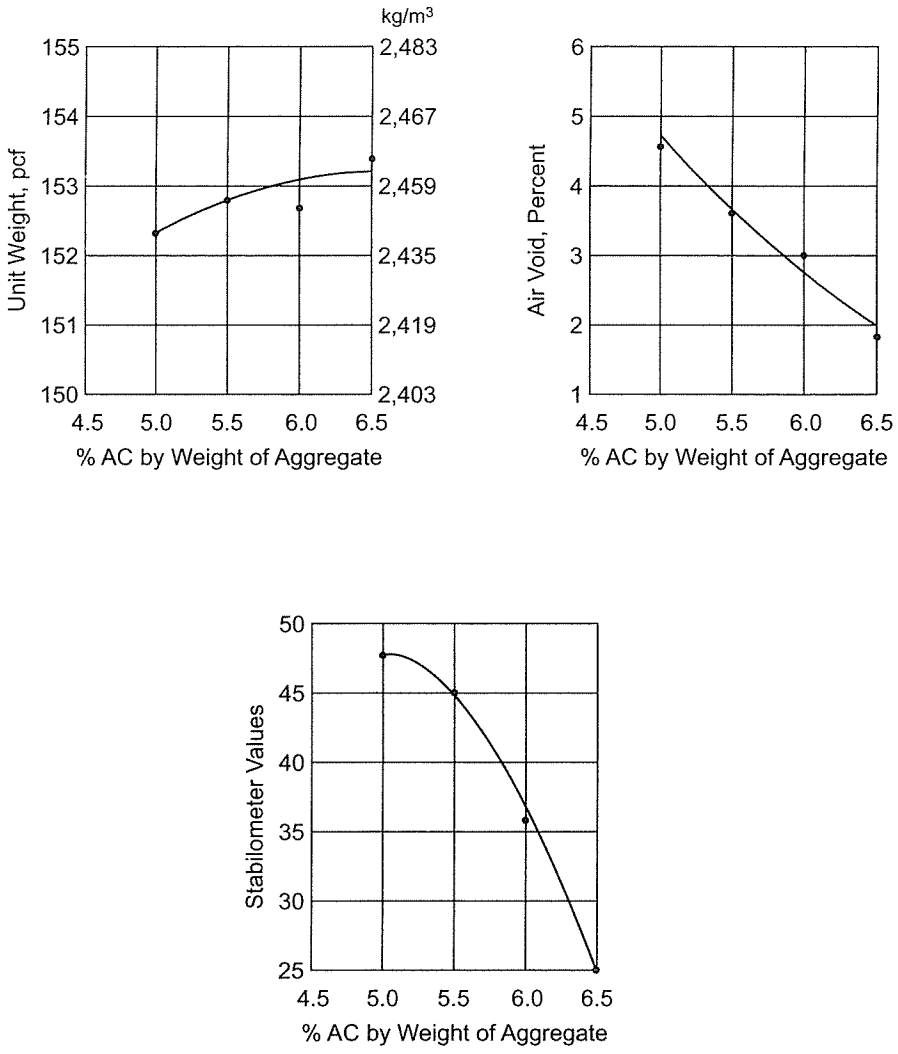


Figure 4.23 Test Property Curves for Hot Mix Design Data Using the Hveem Method

Table 4.7
Hveem Design Criteria

Traffic Category	Heavy		Medium		Light	
Test Property	Min.	Max.	Min.	Max.	Min.	Max.
Stabilometer Value	37	–	35	–	30	–
Swell	less than 0.030 in (0.762 mm)					

Notes

1. Although not a routine part of this design method, an effort is made to provide a minimum of approximately 4 percent air-voids.
2. All criteria, and not stability value alone, must be considered in designing an asphalt paving mix.
3. Hot mix asphalt bases that do not meet these criteria when tested at 140°F (60°C) are satisfactory if they meet the criteria when tested at 100°F (38°C) and are placed 4 in (100 millimeters) or more below the surface.
   This recommendation applies only to regions having a range of climatic conditions similar to those prevailing throughout most of the United States. A different lower test temperature may be considered in regions having more extreme climatic conditions.
4. Traffic classifications:
   Light—traffic conditions resulting in a Design ESAL < 10^4
   Medium—traffic conditions resulting in a Design ESAL between 10^4 and 10^6
   Heavy—traffic conditions resulting in a Design ESAL > 10^6

4.6 RAP Mix Design

Hot mix recycling consists of combining reclaimed (or recycled) asphalt pavement (RAP) with new or "virgin" aggregates, and new asphalt binder and/or a recycling agent in a central plant to produce a new HMA mix. The amount of RAP used in an HMA mix is related to the plant type being used, the gradation of the RAP aggregates, and the properties of the asphalt binder in the RAP. The ratio of RAP to virgin aggregates will vary. It is typically 15 to 25 percent for batch plants, and up to 30 to 50 percent for drum plants. The manufacture of an HMA mix using RAP is discussed in detail in Chapter 7, "Manufacture of Hot Mix Asphalt." This section will discuss the mix design procedures. The procedures outlined below are based on AASHTO M 323, *Superpave Volumetric Mix Design,* and the NCHRP report *Recommended Use of Reclaimed Asphalt in Superpave Mix Design Method: Guidelines.*

4.6.1 Materials Evaluation

The first step in the evaluation of the materials for a project that will include RAP in the HMA is to obtain a representative sample of the RAP to be used. A major concern with the use of RAP is the variability of the material. This is because the RAP may come from different layers in the pavement structure or from stockpiles of materials from different sources. Good stockpiling practices or processing the RAP by crushing and/or screening will reduce the variability. RAP can be sampled from the roadway (by coring or bulk samples), from a stockpile, or from the haul trucks. Good sampling practices should be followed. Sampling methods are discussed in detail in Chapter 10.

Aggregate Gradation

The properties of the aggregate after the asphalt has been extracted from the RAP should be determined. The ignition oven (AASHTO T 308 or ASTM D 6307) can be used to determine the RAP composition (asphalt content and gradation) if further testing of the asphalt binder will not be accomplished. If the extracted asphalt binder is to be recovered and tested for future evaluation, AASHTO T 164 or ASTM D 2172 should be used for the extraction process. After extraction of the asphalt binder from the mix, the gradation of the aggregate is determined. Any deficiency in the gradation can be corrected by blending the appropriate fractions of virgin and reclaimed aggregate to meet the gradation requirements of the final HMA mixture. It is also recommended that the angularity of the fine and coarse aggregates be determined. This information may influence the percentage of RAP to be used for a particular project.

To calculate the VMA, it is necessary to determine the combined bulk specific gravity of the aggregates (RAP and virgin aggregates). Two approaches are used. The first approach consists of splitting the extracted aggregate into fine (minus No. 4 [4.75 millimeters]) and coarse fractions (plus No. 4 [4.75 millimeters]), and determining the specific gravity of each fraction. The advantage is that this provides a direct measure of the specific gravity of the aggregate. The disadvantages are that the extraction process (chemical or ignition) will damage the aggregate or that the samples will contain asphalt or solvent, which will make it difficult to accurately measure the specific gravity. The second approach consists of determining the maximum specific gravity of the RAP (G_{mm}) using AASHTO T 209 or ASTM D 2041, determining the asphalt content of the RAP, and then calculating the effective specific gravity (G_{se}) of the aggregate. The advantage of this method is that it is more accurate than the determination of the bulk specific gravity of the individual fractions. The disadvantage is that using G_{se} rather than G_{sb} of the RAP aggregate will result in an overestimate of the combined bulk specific gravity and will artificially raise the VMA of the HMA mix. The error will increase with higher percentages of RAP.

Asphalt Properties

The extracted asphalt obtained using AASHTO T 164 or ASTM D 2172 is recovered from the extract solution using either AASHTO T 170 or ASTM 1856, *Recovery of Asphalt from Solution by Abson Method,* or by ASTM D 5404, *Recovery of Asphalt from Solution Using the Rotavapor Apparatus.* The Rotavapor procedure is the preferred method because it results in less severe changes in the binder properties.

This recovered binder is used to construct blending charts and to characterize the high-, intermediate-, and low-temperature properties of the binder. Approximately 50 grams of recovered binder is needed to conduct these tests. The testing should include dynamic shear rheometer (DSR) tests on the recovered asphalt before and after aging in the rolling thin film oven (RTFO). Bending beam rheometer (BBR) tests should also be conducted on the recovered asphalt after aging in the RTFO.

The DSR test on the recovered binder will be used to determine the *critical high temperature* T_c (High) where $G^*/\sin\delta$ = 1.00 kiloPascals. This is calculated using the following procedure:

Determine the slope of the stiffness-temperature curve as follows:

$$\alpha = \frac{\Delta \log(G^* \sin\delta)}{\Delta T}$$

Determine T_c (High) to the nearest 0.1°C using the following equation:

$$T_c\,(High) = \left(\frac{Log(1.00) - Log(G_1)}{\alpha}\right) + T_1$$

where

G_1 = the $G^*/\sin\delta$ value at a specific temperature T_1 and

α = the slope determined from the above equation

The DSR test on the recovered binder *after* RTFO aging will be used to determine the critical high temperature T_c (High) where G*/sinδ = 2.20 kiloPascal. This is calculated using the following procedure:

Determine the slope of the stiffness-temperature curve as follows:

$$\alpha = \frac{\Delta \log\left(\frac{G^*}{\sin\delta}\right)}{\Delta T}$$

Determine T_c (High) to the nearest 0.1°C using the following equation:

$$T_c\,(High) = \left(\frac{Log(2.20) - Log(G_1)}{\alpha}\right) + T_1$$

where

G_1 = the G*/sinδ value at a specific temperature T_1 and
α = the slope determined from the above equation

The above information is used to determine the critical high temperature T_c (High) of the recovered binder. This is defined as the lowest of the original DSR and RTFO DSR critical temperatures. The high-temperature performance grade of the recovered RAP binder is based on this single critical high temperature.

The next step is to determine the *critical intermediate temperature* T_c (Int). This is determined from DSR testing on RTFO-aged recovered RAP binder, which is used to determine the critical intermediate temperature T_c (Int) as though the RAP binder were pressure-aging vessel (PAV) aged. This is calculated using the following procedure:

Determine the slope of the stiffness-temperature curve as follows:

$$\alpha = \frac{\Delta \log(G^* \times \sin\delta)}{\Delta T}$$

Determine T_c (Int) to the nearest 0.1°C using the following equation:

$$T_c(Int) = \left(\frac{Log(5000) - Log(G_1)}{\alpha}\right) + T_1$$

where

G_1 = the G*/sinδ value at a specific temperature T_1 and
α = the slope determined from the above equation

The last step is to determine the *critical low temperature* T_c(S) or T_c(m) based on the results of performing a BBR testing of the RTFO-aged recovered RAP binder. It is based on the bending beam rheometer (BBR) stiffness, or m-value. T_c(S) is calculated using the following procedure:

Determine the slope of the stiffness-temperature curve as follows:

$$\alpha = \frac{\Delta \log(S)}{\Delta T}$$

Determine $T_c(S)$ to the nearest 0.1°C using the following equation:

$$T_c(S) = \left(\frac{Log(300) - Log(S_1)}{\alpha}\right) + T_1$$

S_1 = the S value at a specific temperature T_1 and
α = the slope determined from the above equation

$T_c(m)$ is calculated using the following procedure:
Determine the slope of the $T_c(m)$ value temperature curve as follows:

$$\alpha = \frac{\Delta m - value}{\Delta T}$$

Determine $T_c(m)$ to the nearest 0.1°C using the following equation:

$$T_c(m) = \left(\frac{0.300 - m_1}{\alpha}\right) + T_1$$

where

m_1 = the m value at a specific temperature T_1 and
α = the slope determined from the above equation

The above information is used to determine the two critical low temperatures, $T_c(S)$ or $T_c(m)$, which are used to represent the critical temperature for the recovered binder $T_c(Low)$. The low-temperature performance grade of the recovered binder is based on this single critical temperature.

4.6.2 Mix Design

Determine Combined Aggregate in Recycle Mix

The combined gradation for the HMA mix using RAP must meet the specifying agency's requirements. To accomplish this, the contractor will blend the aggregates as discussed in Chapter 3. The gradation to be used for this analysis should be the extracted aggregate gradation.

Asphalt Demand of Combined Aggregate

The asphalt demand for the HMA mix using RAP will be determined using one of the standard design procedures: Superpave, Marshall, or Hveem. The final asphalt content will consist of the asphalt in the RAP and the virgin asphalt to be added during the production process. A major advantage of the use of RAP is that it reduces the need for new, virgin asphalt binder.

Selecting the Grade of New Asphalt

The grade of the new asphalt depends on the percentage of RAP to be used. AASHTO M 323 provides the following guidelines for RAP mixtures; see Table 4.8

Table 4.8
Binder Selection Guidelines for RAP Mixtures

Recommended Virgin Asphalt Binder Grade	RAP Percentage
No change in binder selection	< 15
Select a virgin binder one grade softer than normal (e.g., select a PG 58–28 if a PG 64–22 would normally be used)	15–25
Select a virgin binder using the blending charts	> 25

Development of Final Mix Design

The final mix design is developed following one of the standard design procedures (Superpave, Marshall, or Hveem) and must meet the requirements of those design procedures.

4.7 Aggregate Packing Analysis Using the Bailey Method

A procedure to evaluate the aggregate packing properties in HMA, called the Bailey Method, is able to predict volumetric and compactability characteristics of HMA. The system was first developed by Robert D. Bailey, who worked with the Illinois District 5 Materials Bureau. Further refinements have been developed by the Heritage Research Group of Indianapolis, Indiana.

The procedure is based on the following four main principles:

1. % PCS—Percent passing the Primary Control Sieve
2. CA Ratio—The Coarse Aggregate Ratio
3. FA_c Ratio—The Fine Aggregate Coarse Ratio
4. FA_f Ratio—The Fine Aggregate Fine Ratio

Percent PCS

The Bailey Method is based on the concept that the coarse (largest) particle sizes of any mixture creates voids that are occupied by finer aggregates, asphalt, and air. The Primary Control Sieve (PCS) is the aggregate size that determines what is coarse and what is fine for any aggregate or aggregate combination. The volume of coarse aggregate (retained on the PCS) determines if a mix is a "coarse graded mix," a "fine graded mix" or a stone matrix asphalt (SMA) mix.

A **coarse graded mix** is defined as mixture that has a volume of coarse aggregate great enough to develop stone-on-stone contact. The voids in the coarse aggregate then determine the amount of fine aggregate that can be accommodated in the mix.

An **SMA mix** is a coarse graded mixture that contains a volume of fine aggregate that is significantly less than the volume of voids created by the coarse aggregate.

A **fine graded mix** is defined as a mixture that contains a volume of fine aggregate that exceeds the volume of voids that exist in the coarse aggregate when stone-on-stone contact is achieved. The volume of voids in the coarse aggregate is determined utilizing methods described in AASHTO T 19.

The Bailey Method has the ability to estimate the change in VMA (voids in the mineral aggregate) based solely on the change in gradation between two samples. It will not calculate VMA prior to laboratory testing. Once a known VMA is determined on a compacted specimen, the amount and direction of change in VMA can be estimated by the Bailey Method based on gradation changes. However, it is extremely important to remember that the rate of change and direction of change is different for each of the three mix types.

CA, FA_c and FA_f Ratios

The CA, FA_c and FA_f ratios are factors used in the Bailey Method that are calculated on specific fractions of the combined aggregate used in HMA. The CA ratio describes the distribution of large coarse aggregate particles and smaller coarse aggregate particles. This distribution of particles retained on the PCS determines the number and sizes of voids created by the coarse fraction of the mix. The FA_c ratio provides a comparison of coarse sand (material passing the PCS) and fine sand. The coarse sand fraction of the mix creates voids within the coarse aggregate structure that are filled by fine sand. The FA_f ratio is very similar to the FA_c ratio but deals with the finest fractions in the mixture. VMA movement and compaction characteristics due to gradation change are influenced, described, and predicted by the CA, FA_c and FA_f ratios.

The Bailey Method is a very powerful tool to analyze VMA and compactability of HMA mixtures that cannot be covered in depth in this manual. Transportation Research Board Circular #E-C044 provides in-depth discussion of the method.